MATH 420-01

2.2 The Matrix Representation of a Linear Transformation

Definition:

Let V be a vector space over a field $\mathbb{F}$. A basis with a fixed ordering on the elements is called an *ordered basis*. We abuse the standard definition of a set from here on in!

Definition:

Let V be a finite dimensional vector space over a field $\mathbb{F}$. Let $\beta = \{u_1, u_2, ..., u_n\}$ be an ordered basis for V. $\forall\, v \in V$, $\exists$ unique $a_1, a_2, ..., a_n \in \mathbb{F}$ such that $v = \sum_{j=1}^{n} a_j u_j$. We define the *coordinate vector of v relative to* β by

$$[v]_\beta = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

EXAMPLE:

Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. Let

$\beta = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$.

$A = 1\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + 2\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + 3\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + 4\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$.

$[A]_\beta = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$.

Arithmetic on vectors works just like you would expect, as long as you fix the same basis throughout.

Definition:

Let V and W be a finite dimensional vector spaces over a field $\mathbb{F}$ with ordered bases $\beta = \{v_1, v_2, ..., v_m\}$ and $\gamma = \{w_1, w_2, ..., w_n\}$ respectively. Let $T : V \to W$ be a linear transformation. The **matrix representation of T relative to β and γ is the matrix** $A = [a_{ij}]$, where

$$T(v_j) = \sum_{i=1}^{n} a_{ij}\, w_i$$

and is denoted by $[T]_\beta^\gamma$.

ie. The j-th column of $[T]_\beta^\gamma = A$ is formed from the coefficients of $T(v_j)$ expressed in the γ basis.

EXAMPLE:

Let $D : \mathbb{P}_3 \to \mathbb{P}_2$. Let $\beta = \{1, x, x^2, x^3\}$ and $\gamma = \{1, x, x^2\}$ be the derivative transformation.

$D(1) = 0 = 0\,(1) + 0\,x + 0\,x^2$

$D(x) = 1 = 1\,(1) + 0\,x + 0\,x^2$

$D(x^2) = 2\,x = 0\,(1) + 2\,x + 0\,x^2$

$D(x^3) = 3\,x^2 = 0\,(1) + 0\,x + 3\,x^2$

$$[D]_\beta^\gamma = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

Discussion:

Let V and W be vector spaces over a field $\mathbb{F}$. We add and scale linear transformations in the standard way. With these two operations, the set of linear transformation from V to W also form a vector space over $\mathbb{F}$, denoted by $\mathcal{L}(V, W)$, (or just $\mathcal{L}(V)$ when $W = V$). Moreover, if $T : V \to W$ and $U : V \to W$ are linear transformations, then for bases β and γ,

(i) $[T + U]_\beta^\gamma = [T]_\beta^\gamma + [U]_\beta^\gamma$

(ii) $[a\,T]_\beta^\gamma = a[T]_\beta^\gamma$

2.3 Composition of Linear Transformations and Matrix Multiplication

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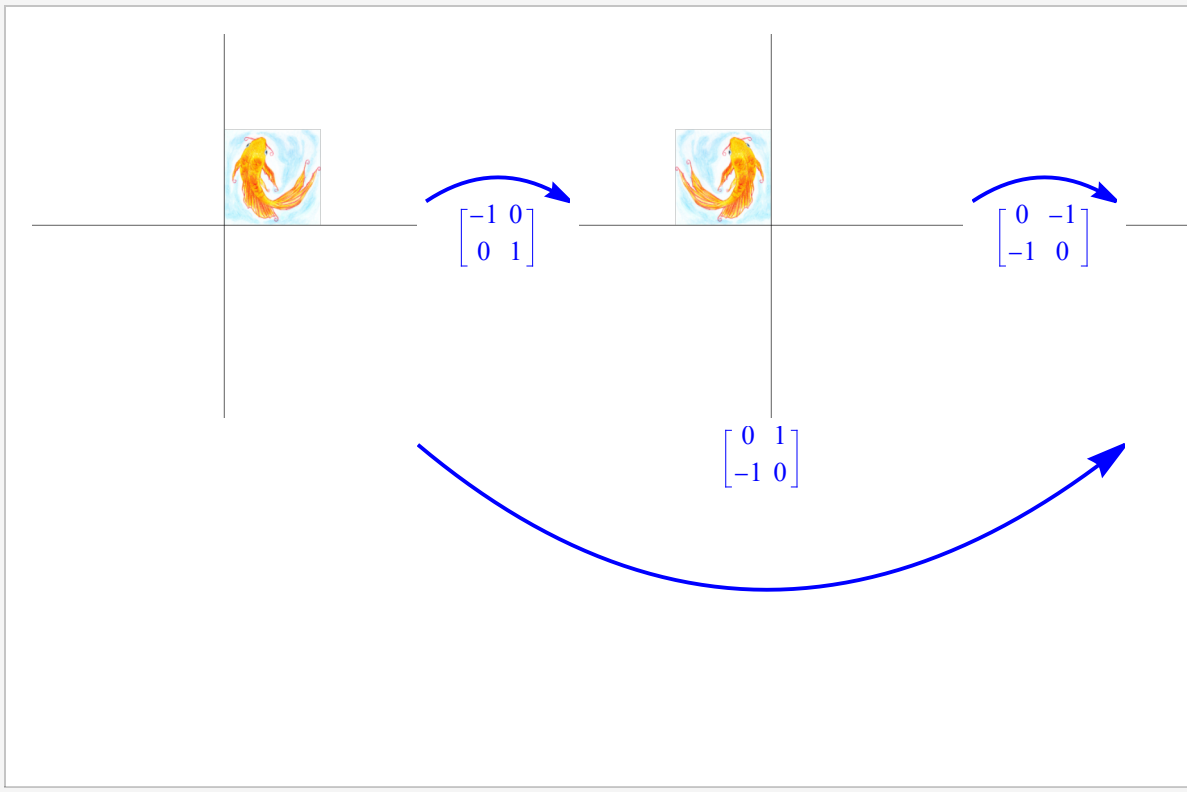

choose first transformation | *reflection in x_2 –axis*

choose second transformation | *reflection in $x_1 = -x_2$*

Matrices as composition of linear transformations.